Milestone – 3 Report

Project Title

MedAssist AI: Medical Symptom Analysis & Disease Prediction System

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1. Introduction

MedAssist AI is an intelligent healthcare application developed to predict diseases based on symptoms entered by users. The objective of this milestone was to integrate the trained Decision Tree machine learning model into a web-based application. The frontend, backend, and machine learning components were successfully connected to create a complete end-to-end disease prediction system.

The application enables users to enter symptoms through a user-friendly interface. These symptoms are processed by the Flask backend, where the trained Decision Tree model predicts the most probable disease. The prediction result is then displayed instantly on the frontend. This milestone demonstrates successful integration of machine learning with full-stack web development.

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2. Objectives

- Integrate the trained Decision Tree model into the web application.
- Develop a REST API using Flask.
- Connect the React frontend with the backend.
- Accept symptom inputs from users.
- Generate disease predictions in real time.
- Validate the complete workflow through testing.
- Build an efficient and user-friendly healthcare prediction system.

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3. Technologies Used

Technology	Purpose
Python 3.12	Programming Language
React.js	Frontend Development
HTML5	Web Structure
CSS3	User Interface Design
JavaScript	Client-side Programming

Technology	Purpose
Flask	Backend Framework
Scikit-learn	Machine Learning
Decision Tree Classifier	Disease Prediction Model
Pandas	Data Processing
NumPy	Numerical Computation
Joblib	Model Serialization
Visual Studio Code	IDE
Git & GitHub	Version Control

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4. Integration Process

The integration process involved loading the trained Decision Tree model using Joblib into the Flask backend. REST API endpoints were developed to receive symptom inputs from the React frontend. The backend converted the symptoms into the required feature vector and passed them to the Decision Tree model. The model generated a disease prediction, which was returned to the frontend in JSON format. Finally, the frontend displayed the predicted disease to the user in a simple and interactive interface.

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5. Experimental Results

The integrated system was tested using multiple symptom combinations to evaluate the functionality of the application.

Machine Learning Model: Decision Tree Classifier

Accuracy: 86%

The Decision Tree model achieved an overall accuracy of **86%**, providing reliable disease predictions based on user symptoms. The application successfully handled user requests, processed inputs, generated predictions, and displayed the results without integration issues.

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6. Features Implemented

- Symptom-based disease prediction
- User-friendly React interface
- Flask REST API integration
- Real-time prediction
- Model loading using Joblib

- Input validation
- Fast response time
- End-to-end application workflow

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7. Challenges Faced

Several challenges were encountered during integration, including establishing communication between the React frontend and Flask backend, loading the serialized Decision Tree model, handling API requests and responses, validating user inputs, and resolving dependency-related issues. These challenges were addressed through systematic testing and debugging.

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8. Learning Outcomes

This milestone enhanced our understanding of full-stack application development, machine learning model deployment, REST API development using Flask, frontend-backend communication, Git version control, and end-to-end system integration. It also provided practical experience in deploying machine learning models within real-world web applications.

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9. Future Enhancements

- Deploy the application on a cloud platform.
- Improve prediction accuracy using advanced ensemble models.
- Integrate medicine recommendation features.
- Add user login and authentication.
- Maintain prediction history.
- Develop a mobile application.
- Integrate healthcare APIs for enhanced functionality.

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10. Conclusion

The MedAssist AI project was successfully integrated using React.js, Flask, and a Decision Tree Classifier. The system provides real-time disease prediction based on symptoms entered by users. The Decision Tree model achieved **86% accuracy**, demonstrating reliable performance for disease prediction. This milestone successfully completed the end-to-end integration of the machine learning model with the web application and established a strong foundation for future deployment and feature enhancements.